UGENDRAN S

Full Stack Engineer | AI-Integrated Systems | Government & Enterprise

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PROFESSIONAL SUMMARY

Product-focused Full Stack Engineer with 3+ years building high-scale government and enterprise platforms. Specialized in AI-powered systems (RAG, LLMs), microservices architecture, and performance-optimized databases handling millions of records. Delivered 10+ production systems across government, banking, healthcare, and education sectors — consistently reducing manual effort by 45–90% through scalable system design, CI/CD pipelines, and workflow automation.

TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Bootstrap
Backend	Node.js, Express.js, NestJS, PHP, CakePHP 5, FastAPI, RESTful APIs
AI / ML	Python, RAG, LLM, Transformer Models, PyTorch, Pandas, NumPy
Databases	PostgreSQL, Oracle DB, MySQL | Partitioning, Indexing, Materialized Views
Payments	Razorpay, Worldline, CCAvenue
DevOps	AWS, Docker, Git, CI/CD, Ubuntu Linux
Architecture	Microservices, System Design, RESTful APIs, Scalable Architecture
AI Tools	GitHub Copilot, Claude AI, ChatGPT, Figma AI, Stitch AI, n8n
Certifications	Generative AI Mastermind – Outskill (2025)

WORK EXPERIENCE

Software Developer — Muthu Soft Labs Pvt Ltd *March 2023 – Present*

- Built 6+ enterprise web applications across government, healthcare, and education using microservices architecture — reducing manual processing time by 60%.
- Integrated Razorpay, Worldline, and CCAvenue payment gateways for 10,000+ users, reducing payment failure rate by 30%.
- Designed system architecture and optimized RESTful APIs and Oracle/PostgreSQL queries for ERP workflows, improving response times by 40%.
- Built RAG-based AI reporting system trained on 50,000+ documents — reduced report generation time by 90% at 95% accuracy.
- Set up CI/CD pipelines and worked directly with government and enterprise clients to deliver production-grade systems on time.

KEY PROJECTS

Academic Resource Management System *Jan 2026 – Present*

Stack: CakePHP 5, MySQL, AJAX, Tailwind CSS | **Sector:** AVIT

- Built timetable scheduling and library management modules for 5,000+ students.
- Implemented dynamic timetable generation with conflict detection, reducing scheduling errors to zero.
- Developed book issuance, return tracking, and fine calculation workflows with real-time availability updates.

Textbook ERP Management System *Mar 2025 – Sep 2025*

Stack: React.js, TypeScript, Node.js, Express.js, Oracle DB | **Sector:** Government

- Built ERP for government textbook procurement, inventory tracking, and distribution — cut procurement cycle time by 45%.

- Designed Oracle DB model managing statewide inventory across 500+ distribution points with real-time stock visibility.
- Built role-based access for district officers and warehouse admins with full audit trail for every stock movement.

Government Survey Dashboard – Big Data Analytics *Feb 2025 – Present*
Stack: Express.js, TypeScript, PostgreSQL, RAG, AI Model | **Sector:** Government

- Built analytics dashboard processing millions of survey records; optimized PostgreSQL with partitioning and materialized views — improved retrieval speed by 80%.
- Integrated RAG-based AI model to auto-generate survey insights, reducing analyst reporting time by 75%.
- Built streaming APIs handling 10M+ records with <300ms response time and drill-down filters by region and time period.

POS Mathi – Product Cart & Payment Platform *Jan 2025 – Present*
Stack: Next.js, NestJS, Tailwind CSS, Oracle DB, Payment Gateway | **Sector:** Government

- Built district and state-level vendor cart and payment platform with role-based access for admins and vendors.
- Integrated payment gateway reducing manual settlement effort by 50%; optimized Oracle DB for high-volume concurrent transactions.
- Designed state-wide reporting dashboards with real-time transaction visibility for government procurement teams.

Urban House Management System *Nov 2024 – Apr 2025*
Stack: React.js, TypeScript, Express.js, PostgreSQL, Payment Gateway | **Sector:** Government

- Built housing platform covering property listings, tenant onboarding, rent collection, and maintenance workflows.
- Integrated payment gateway for automated rent billing, reducing manual collection effort by 70%.
- Designed PostgreSQL schema with optimized indexing, achieving sub-200ms average API response times.

College Account & Payment Management System *Aug 2024 – Mar 2025*
Stack: CakePHP 5, MySQL, CCAvenue | **Sector:** AVIT

- Built fee demand, refund, and CCAvenue payment system processing 10,000+ transactions per year.
- Automated fee demand generation with due-date reminders, reducing manual follow-up effort by 60%.
- Implemented payment reports and receipt generation for finance department and student self-service portal.

COE – Controller of Examinations System *May 2024 – Aug 2024*
Stack: CakePHP 5, MySQL, AJAX, Tailwind CSS | **Sector:** Education

- Delivered exam registration, hall ticket generation, and result publication system for 15,000+ students.
- Automated hall ticket and result workflows — cut admin turnaround from days to hours.
- Built role-based access for exam controllers, faculty, and students with real-time status tracking.

Christian Welfare Claim System *Jan – May 2024*
Stack: React.js, TypeScript, Tailwind CSS, Node.js, MySQL | **Sector:** Government

- Built multi-level welfare claim approval platform with role-based dashboards, reducing approval cycle time by 55%.
- Implemented document upload, verification, and status tracking for beneficiaries and approving authorities.
- Designed MySQL schema supporting hierarchical approval chains across district, state, and central levels.

IOB Cheque Processing System *Mar – Dec 2023*
Stack: React.js, TypeScript, Express.js, IOB Bank API | **Sector:** Banking

- Built real-time cheque validation platform integrated with IOB Bank third-party API, eliminating manual entry errors.
- Developed React.js TSX operator dashboard with live status tracking for bank operators.
- Designed secure Express.js backend handling API communication with IOB, reducing cheque processing time significantly.

NHM – National Health Management System *Mar – Aug 2023*

Stack: CakePHP 5, PostgreSQL, AJAX, Tailwind CSS | **Sector:** Government

- Built health scheme management and beneficiary tracking system serving 50,000+ users across districts.
- Implemented eligibility verification, application submission, and approval workflows for health officers.
- Designed PostgreSQL schema with optimized queries for reporting beneficiary data across multiple health programmes.

EDUCATION

B.E. – Electronics & Communication Engineering *2019 – 2023*

NSN College of Technology, Karur